Table 2 Metric-specific deviations introduced by wrist placement

Summary overview of metric errors due to wrist wearing position, compared to glasses	
Errors are all based on their modelling scale	
	Wrist
Duration	**2%** (0% - 8%) **signif 8/12**, +2/-10
Exposure history	**1%** (0% - 1%) **signif 0/2**, +2/-0
Level	**9%** (1% - 84%) **signif 7/9**, +2/-7
Spectrum	**5%** (5% - 5%) **signif 1/1**, +0/-1
Temporal dynamics	**15%** (5% - 52%) **signif 7/7**, +6/-1
Timing	**2%** (0% - 9%) **signif 4/23**, +21/-2
Overall	**2%** (0% - 84%) **signif 27/54**, +33/-21
Overall (significant)	**8%** (1% - 84%) +11/-16
median error (min - max), significant errors / all metrics, positive / negative deviations	

Errors due to wrist placement, compared with glasses-based measurement, vary across exposure summaries. Some quantities show small median deviations, while others, particularly level and temporal dynamics, show larger deviations or wider ranges. This supports a construct-specific validation strategy: wrist-worn light measures should be evaluated according to the metric and hypothesis for which they are used, rather than treated as uniformly valid or invalid. Data adapted from Zauner et al., “Sensor placement causes outcome-dependent bias in ambulatory light-exposure estimates” (bioRxiv, 2026; https://doi.org/10.64898/2026.07.28.741277). Abbreviations: max, maximum; min, minimum; signif, significant